

Incoming SCADA Telemetry Stream

Clean Data | Buffer Delay $\tau \in [10, 60]$ min | Pitch Withheld ($\beta = \emptyset$)



Observability & Staleness Detector

Verifies pitch register $\exists \beta$ and timestamp staleness τ

Clean: $\tau = 0, \exists \beta$

$\tau \leq 20$ min, $\exists \beta$

$\beta = \emptyset$ or $\tau \rightarrow 60$ m

Tier 1: Continuous Physical Quantile

- Nominal $\tau = 0$
- 589.5k kW·h
- 6.8% violation
- Zero ML compute
- Deterministic rule

[Physics Optimal]

Tier 2: State-Cond. Recalibration

- $\tau \in [10, 20]$ min
- Full pitch β
- Absorbs 55.3% shortfall loss
- Empirical $q_{\tau}(v)$

[Recalibration]

Tier 3: STGQ Modular Residual Head

- $\beta = \emptyset$ or $\tau = 60$
- 46.7k--68.6k saving
- 9.7% violation
- Recovers latent operating state

[ML Representation]